Subtleties of the General Binomial Expansion

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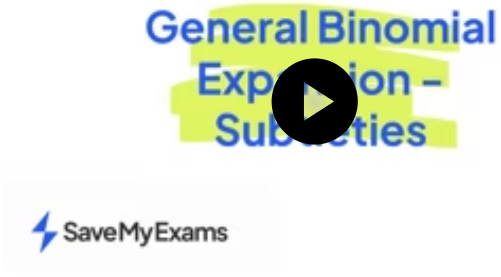

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Help me understand

Subtleties of the general binomial expansion

What is the general binomial expansion?

- The binomial expansion applies for positive integers, $n \in \mathbb{N}$
- The **general binomial expansion** applies to other types of powers too

THE GENERAL BINOMIAL THEOREM

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

FOR $n \in \mathbb{R}$ AND $|x|<1$

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- The **general binomial expansion** applies for all **real** numbers, $n \in \mathbb{R}$
- Usually fractional and/or negative values of **n** are used
- It is derived from $(a + b)^n$, with $a = 1$ and $b = x$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How do I use the general binomial expansion to expand $(a+kx)^n$?

The general binomial expansion can be applied to expanding $(\mathbf{a + kx})^{\mathbf{n}}$

STEP 1 Rewrite the question into $(\mathbf{1 + bx})^{\mathbf{n}}$ form

- Roots are fractional powers
- Denominators are negative powers
- A factor may be needed to make "**a = 1**"

STEP 2 Replace "x" by "bx" in the expansion

Check carefully to see if **b** is negative

STEP 3 Expand and simplify

Use a line for each term to make things easier to read and follow

Use brackets

STEP 4 If required, check and state the validity statement, **|bx| < 1**

e.g. FIND THE FIRST 3 TERMS IN THE EXPANSION OF $(3+2x)^{-4}$ AND STATE THE VALUES OF x FOR WHICH IT IS VALID.

STEP 1: $(3+2x)^{-4} = [3(1+\frac{2}{3}x)]^{-4}$

$= 3^{-4}(1+\frac{2}{3}x)^{-4}$ ← $(1+bx)^n$ FORM

STEP 2: $= \frac{1}{81}[1 + (-4)(\frac{2}{3}x) + \frac{(-4)(-5)}{2}(\frac{2}{3}x)^2 + \ldots]$

REPLACE "x" WITH "$-\frac{2}{3}x$"

STEP 3: $= \frac{1}{81}(1 - \frac{8}{3}x + \frac{40}{9}x^2 + \ldots)$

EXPAND AND SIMPLIFY

$\therefore \quad (3+2x)^{-4} \approx \frac{1}{81} - \frac{8}{243}x + \frac{40}{729}x^2$

STEP 4: $|\frac{2}{3}x| < 1$

$|x| < \frac{3}{2}$

"$|bx| < 1$"

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Worked Example

the expansion of $\dfrac{1}{\sqrt[3]{8-3x}}$

(b) Write down the values of x for which the expansion in part (a) is valid.

a) STEP 1:

$$\frac{1}{\sqrt[3]{8-3x}} = (8-3x)^{-\frac{1}{3}}$$
$$= \left[8\left(1-\frac{3}{8}x\right)\right]^{-\frac{1}{3}}$$
$$= 8^{-\frac{1}{3}}\left(1-\frac{3}{8}x\right)^{-\frac{1}{3}}$$

REWRITE IN $(1+bx)^n$ FORM

REPLACE "x" WITH "$-\frac{3}{8}x$"

CARE! $-\frac{1}{3}-1=-\frac{4}{3}$

STEP 2:

$$= \frac{1}{\sqrt[3]{8}}\left[1+\left(-\frac{1}{3}\right)\left(-\frac{3}{8}x\right)+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)}{2}\left(-\frac{3}{8}x\right)^2+\ldots\right]$$

STEP 3:

$$= \frac{1}{2}\left(1+\frac{1}{8}x+\frac{1}{32}x^2+\ldots\right)$$

$$\therefore \frac{1}{\sqrt[3]{8-3x}} \approx \frac{1}{2}+\frac{1}{16}x+\frac{1}{64}x^2$$

EXPAND AND SIMPLIFY

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b) STEP 4:

$$\left|-\frac{3}{8}x\right| < 1$$

$$\therefore |x| < \frac{8}{3}$$

"$|bx| < 1$"

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